

CaseMatch V1.0.0 User Guide

1 Introduction

CaseMatch is a zero-code, open-source content-based image retrieval (CBIR) software designed for users without programming backgrounds, particularly clinicians and medical researchers. It enables the construction of searchable medical image libraries using state-of-the-art visual encoders through an intuitive graphical interface. The software employs ResNet50 as the core visual feature extractor and provides four pretrained checkpoints, all accessible via a unified GUI.

2 Configuration Files

The software utilizes two configuration files: `config.json` and `model_config.json`.

The `config.json` file contains the following parameters:

```
{
  "common": {
    "batch_size": 16,
    "input_size": 224,
    "resize_before_crop": 256
  }
}
```

The `input_size` and `resize_before_crop` parameters follow the default settings for ResNet50. For smaller images, such as 224×224 H&E histopathology patches, `resize_before_crop` can be adjusted to 224 to avoid unnecessary upscaling.

The default `batch_size` of 16 is optimized for stable execution on both CPU and GPU environments. Users may adjust this value based on available hardware resources.

The `model_config.json` file manages the four integrated ResNet50 checkpoints. The software dynamically loads models by reading this configuration file. Users can incorporate additional supervised or self-supervised domain-specific checkpoints by updating the entries in this file, without modifying the source code.

3 Retrieval Modes: Fast vs. Full

CaseMatch provides two retrieval modes to accommodate different clinical workflows, balancing speed and interpretability.

3.1 Fast Mode (Embedding-Only)

Fast mode performs content-based retrieval using only the deep feature embeddings, without generating Grad-CAM visualizations. This is the **default** mode and is recommended for:

- Rapid case browsing during busy clinical sessions
- Large-scale screening where interpretability is not immediately required
- Situations where minimal latency is critical

Performance: On the Kvasir-v2 gallery (6,400 images), fast mode completes in approximately 0.3s on CPU.

Operation: Ensure the `Enable Grad-CAM` checkbox is **unchecked** in the Image Retrieval panel before clicking `Start Retrieval`. The software returns the top-*k* most similar images with their cosine similarity scores. No heatmaps or overlay figures are generated.

3.2 Full Mode (With Grad-CAM)

Full mode extends fast mode by generating Grad-CAM activation maps for both the query image and all retrieved results. This mode is recommended for:

- Educational demonstrations and case discussions
- Research analysis requiring visual validation of similarity regions
- Quality assurance and diagnostic retrospection

Performance: On the Kvasir-v2 gallery (6,400 images), full mode completes in approximately 3s on CPU.

Operation: Check the `Enable Grad-CAM` checkbox in the Image Retrieval panel before clicking `Start Retrieval`. The software generates:

- Query image CAM and overlay heatmaps
- Individual CAM, heatmap, and overlay images for each top- k result
- A consolidated `comparison_overview.png` figure

Users may switch between modes on a per-query basis without rebuilding the index. The in-memory vector cache is shared across both modes, ensuring that project loading time is incurred only once per session.

4 Building an Image Retrieval Library

Creating an image retrieval library involves the following straightforward steps:

1. **Prepare the image collection:** Organize target images in a folder. The software recursively scans all subdirectories to locate supported image files.
2. **Initialize the project:** In the **Build Project** panel, assign a project name, select an available checkpoint, and specify the source image directory.

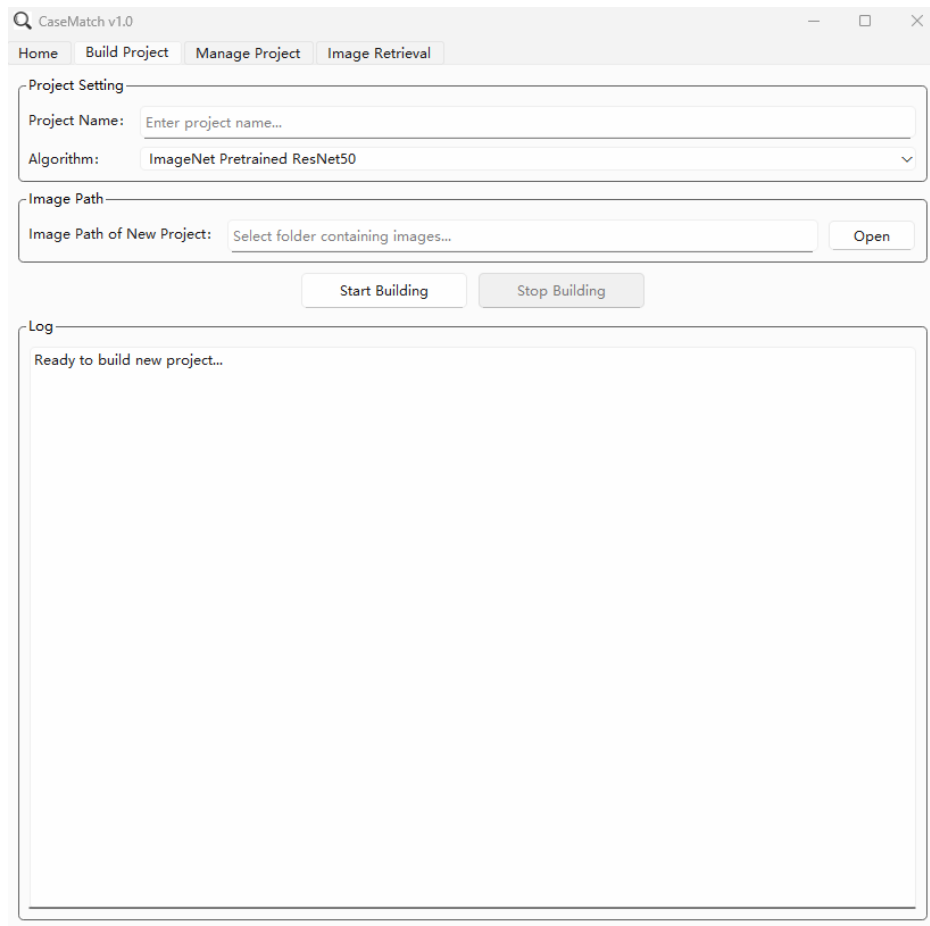


Figure 1. Build Project page.

3. **Vector generation:** The software automatically extracts deep features and constructs the vector index. All metadata and embeddings are stored in the `results` directory under a newly created project folder.

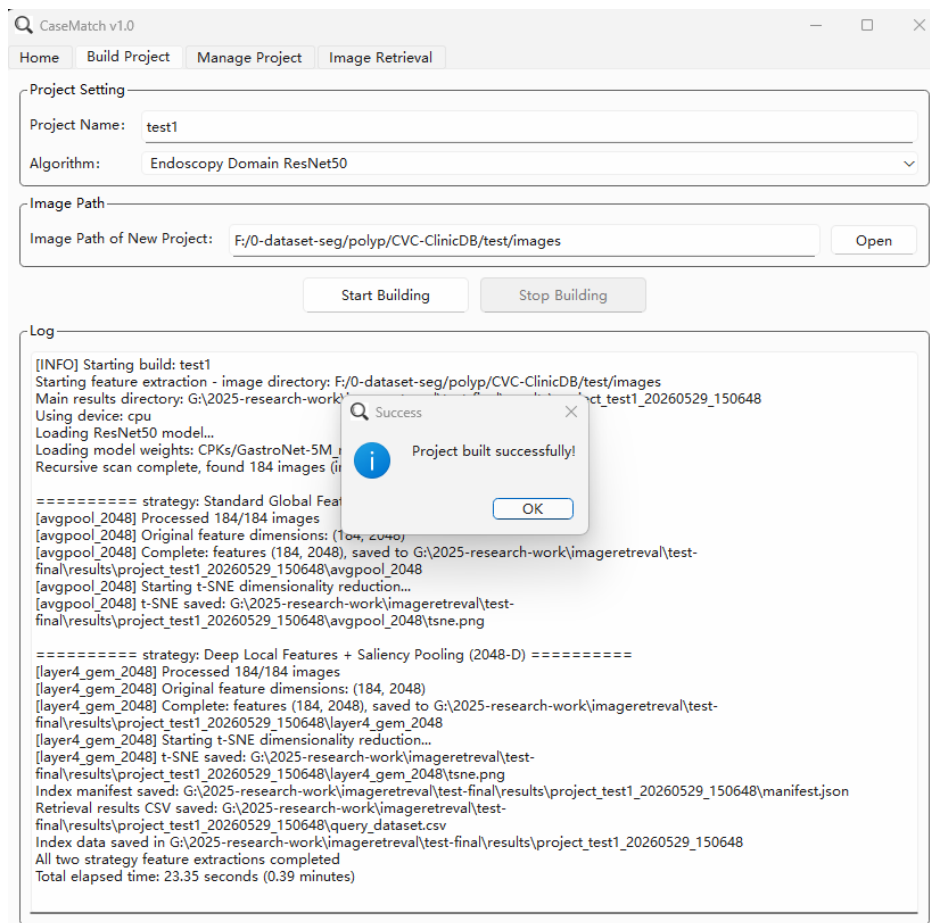


Figure 2. Successful construction of the image vector index library.

- Incremental maintenance:** The source image directory remains fixed after initial construction. Users may add or remove images in this directory. In the **Manage Project** panel, select an existing project and trigger a quick rebuild. The system detects newly added or deleted images and updates the index incrementally. Previously indexed images are not reprocessed, ensuring efficient maintenance.

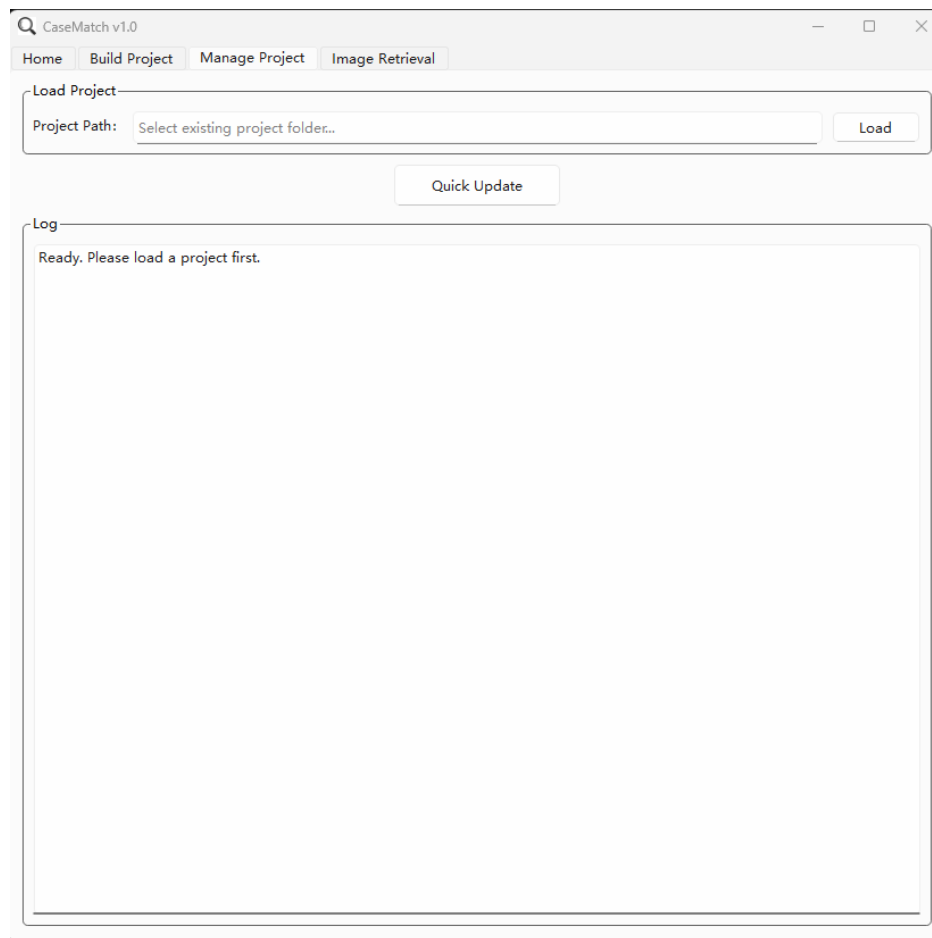


Figure 3. Incremental management for any existing project.

5. **Image retrieval:** In the **Image Retrieval** panel, select an active project, specify a query image path, configure the Top- k retrieval count, and choose the retrieval method. Execute the search to display the Top-5 results in the GUI. Complete retrieval results, including similarity scores and Grad-CAM heatmaps, are exported to the `results` folder.

5.1 Project Management Files

Each project folder stores the configuration file and computed vector embeddings, as shown below:

 avgpool_2048	2026/5/29 15:07
 layer4_gem_2048	2026/5/29 15:07
 manifest.json	2026/5/29 15:07
 project_config.json	2026/5/29 15:07
 query_dataset.csv	2026/5/29 15:07
 training.log	2026/5/29 15:07

Figure 6. Project result directory containing configuration and vector index files.

5.2 Retrieval Outputs

Retrieval results are organized in a dedicated output folder, as illustrated below:

 gradcam	2026/5/29 15:10
 input_image.png	2020/10/20 11:19
 retrieval_20260529_151023.log	2026/5/29 15:10
 retrieval_results.csv	2026/5/29 15:10

Figure 7. Retrieval result directory structure.

The `gradcam` subdirectory contains detailed activation information for each Top- k retrieved image:

 top1_93	2026/5/29 15:10
 top2_94	2026/5/29 15:10
 top3_86	2026/5/29 15:10
 top4_96	2026/5/29 15:10
 top5_90	2026/5/29 15:10
 comparison_overview.png	2026/5/29 15:10
 query_heatmap.jpg	2026/5/29 15:10
 query_overlay.jpg	2026/5/29 15:10

Figure 8. Grad-CAM activation maps for individual retrieved images.

The `comparison_overview` directory provides a consolidated visualization combining the query image, retrieved similar cases, and their corresponding Grad-CAM activation maps:

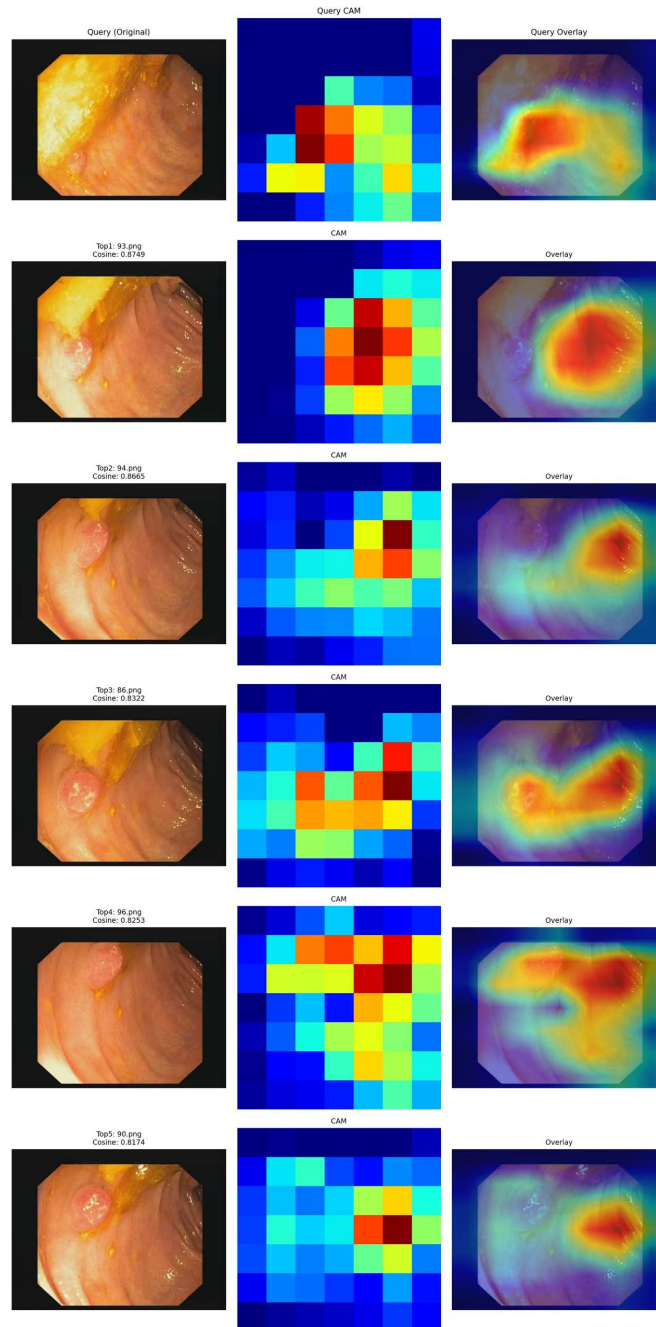


Figure 9. Comprehensive comparison overview showing the query image alongside retrieved similar cases and their Grad-CAM heatmaps.

6 Important Notes

1. **Vector storage:** The software maintains image embeddings in NumPy (.npy) format, along with corresponding file lists and original image paths. Each image is represented as a 2048-dimensional feature vector extracted from ResNet50.
2. **Image path integrity:** The software does **not** store original image files. Users must ensure the stability of source image paths and filenames. Any renaming of image files or restructuring of directory paths will invalidate existing references, necessitating a full reconstruction of the retrieval library.
3. **Extensible model configuration:** Users can extend the software's capabilities by updating the `model_config.json` file to register new ResNet50-based visual extractors. Future releases will integrate additional domain-specific foundation

models (e.g., for X-ray and CT imaging) as validated pretrained weights become publicly available.

7 Conclusion

All code is open-source and licensed under the Apache 2.0 license. If you have any questions or need further assistance, feel free to contact me at weihaomeva@163.com. For those who are interested in custom development or support, I am available for paid consultations.

The code repository is available at: <https://github.com/AI-thpremed/CaseMatch>